IndoorScene-CoT: Enhancing 3D Scene Synthesis via Chain-of-Thought Reasoning

A Dataset Analysis

We conduct a statistical analysis of the dataset. As shown in Figure 1, prominent words include "functionality", "user", "design", "color", "material", and "object". These words highlight key considerations and elements in the scene generation process, such as ensuring functionality, meeting user requirements, and focusing on design aspects like color and material.

Figure 1: Word cloud generated based on the content of the dataset.

We also draw a bar chart to illustrate the frequency of different furniture items in our dataset. As reported in Figure 2, the most frequent items are "rug" and "chair", with "rug" appearing 218 times and "chair" appearing 170 times. Other items like "floor_lamp", "mirror", "side_table", and "plant" have relatively lower frequencies, ranging from 53 to 77.

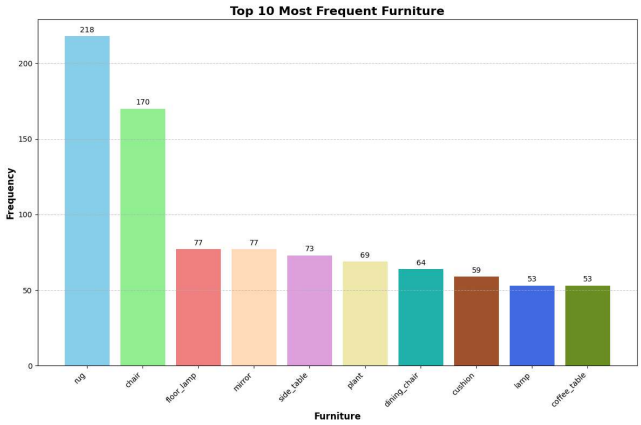

Figure 2: Number of object occurrences in IndoorScene-CoT.

B Dataset Augmentation

Our proposed chain-of-thought (CoT) framework can augment existing indoor scene datasets with explicit reasoning processes, thereby improving their versatility and suitability for training interpretable and spatially aware 3D Scene Synthesis models. Taking the widely used 3D-FRONT dataset [4] as a representative example, we first extract fine-grained attributes for each furniture based on the descriptions provided by ReSpace [2], including visual properties such as color, material, and style. For each room layout containing distinct furniture configurations, we then leverage large language models to generate diverse and semantically rich natural language descriptions as conditional inputs.

Employing Qwen3-VL-8B-Instruct [1] integrated with our cascading framework, we feed the generated textual descriptions and ground-truth layouts into the system. The model decomposes the instructions step-by-step, partitions the indoor space into functional zones, recommends appropriate objects aligned with the prompt, and refines the relative positions and orientations of all objects under the supervision of the layout. In this way, our framework produces complete, human-interpretable chain-of-thought traces that link language instructions to fine-grained spatial layout decisions, yielding 13,484 samples that contains paired text, layout, and reasoning annotations. As shown in Table 1, the same model trained with supervised fine-tuning (SFT) on this augmented 3D-FRONT [4] achieves strong performance on KID (0.0169) and OOR (0.1661), which validates that our framework can effectively enrich existing datasets.

Table 1: Performance on 3D-FRONT [4] augmented with IndoorScene-CoT.

Methods	FID↓	KID↓	OOB↓	OOR↓	GPT-4o Criteria↑
LayoutGPT [3]	106.65	0.0232	0.1926	0.2745	5.20
OpenAI GPT-5.2 [5]	94.11	0.0183	0.0181	0.2215	5.24
Ours with 3D-FRONT [4]	93.37	0.0169	0.2084	0.1661	5.37
Ours	85.31	0.0035	0.0320	0.1759	5.57

C User Study

We conduct a user preference evaluation by anonymously packaging results generated by LayoutGPT [3], HOLODECK [6], and our model, asking users to choose the one they perceive as most effective. Out of 142 responses received, 137 users (96.5%) find the scenes generated by our model more realistic and aesthetically pleasing, 5 users (3.5%) prefer HOLODECK, and no users select LayoutGPT. The results indicate that scenes generated by our model align with real human preferences and design standards in practice.

D Complete Reasoning Process of the Model

D.1 Chain-of-Thought Demonstration

We present a comprehensive demonstration of our trained model's workflow, from user input to the generation of the final scene graph. This includes detailed insights into the model's reasoning processes at each step and the iterative placement of objects, as shown in Figure 10 and Figure 11. Our input text is "A functional home office space equipped with a wooden desk, a leather upholstered swivel chair, and a black metal filing cabinet".

D.2 Visualization

By integrating with the local asset library, we render the generated scene graph to obtain images from four complementary camera views, which are shown in Figure 3.

Figure 3: Visualization of the generated scene graph of our model.

E More Visual Results

We present additional visualizations to demonstrate that our method can generate diverse and complex scenes that align with human aesthetics.

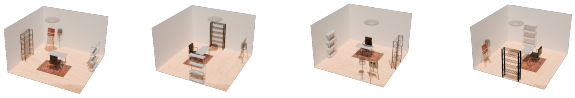

Figure 4: A bright artist's studio with an easel, a canvas storage rack, and a drafting table.

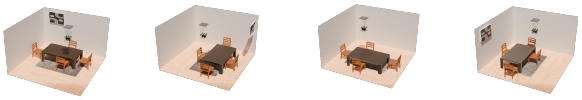

Figure 5: A mid-century modern dining room with a table, a set of chairs, and a pendant light.

Figure 6: A minimalist living room with a sectional sofa, a coffee table, and a floor lamp for ambient lighting.

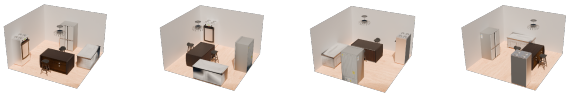

Figure 7: A modern kitchen featuring a stainless steel refrigerator, an oven, and a ceramic sink counter.

Figure 8: A playful children's playroom with a toy storage unit, a child-sized table, and a set of colorful chairs.

Figure 9: A luxurious bathroom with a ceramic white bathtub, a marble sink, and a metal towel rack.

User Preference Analysis: The user envisions a functional home office that incorporates specific furniture pieces such as a wooden desk, a leather upholstered swivel chair, and a black metal filing cabinet. The room measures 5.0 meters by 5.0 meters with a height of 3.0 meters, providing ample space for these items. The primary focus is on creating a workspace that is both efficient and aesthetically pleasing, with additional elements like a rug, wall art, and a plant suggested to enhance the room's warmth and comfort. The user also expressed interest in adding a desk lamp and a monitor to support work activities.

Functional Zone Partition: The room is divided into several functional areas based on the user's requirements. The central area is designated for the desk and chair setup, forming the core of the home office. Adjacent to this is the filing cabinet area, which provides storage while maintaining accessibility. The east wall is reserved for wall art and a plant, adding aesthetic value and warmth to the space. Additional elements like a rug and a desk lamp are strategically placed to enhance the overall functionality and aesthetic of the room.

Object Recommendation: For the central workspace, a classic wooden desk and a leather upholstered swivel chair are recommended to create a cohesive and functional setup. A modern black metal filing cabinet is suggested for storing office supplies. To enhance the ambiance, contemporary wall art and a plant are proposed for the east wall. A minimalist grey rug is recommended to add texture and warmth under the desk area. A modern desk lamp and a monitor are also included to improve lighting and support work activities.

Object Placement: The wooden desk, a central element of the home office, is placed against the north wall, facing the south wall. This placement maximizes space usage and creates a functional workspace, aligning with the user's preference for a classic style. The desk's dimensions are 1.8 meters by 0.9 meters by 0.75 meters, fitting comfortably against the wall and leaving ample space for other furniture.

The leather upholstered swivel chair is positioned directly in front of the desk, facing the north wall. This placement ensures ergonomic seating and complements the desk's orientation, maintaining balance and proportion in the room. The chair's dimensions are 0.7 meters by 0.7 meters by 1.2 meters, allowing it to fit seamlessly in front of the desk without spatial conflicts.

The black metal filing cabinet is placed against the west wall, facing the east wall. This location ensures accessibility from the desk without obstructing the chair's movement, maintaining an open and functional workspace. The filing cabinet's dimensions are 0.8 meters by 0.6 meters by 1.8 meters, fitting well along the wall and complementing the desk and chair arrangement.

The wall art is centrally placed on the east wall, facing the west wall. This placement avoids spatial conflicts with existing furniture and adds visual interest without cluttering the room. The wall art's dimensions are 1.2 meters by 0.05 meters by 0.8 meters, making it suitable for wall display.

The plant is placed on the floor against the east wall, facing the west wall. This placement enhances the room's aesthetic without obstructing functional areas. The plant's dimensions are 0.5 meters by 0.5 meters by 1.5 meters, ensuring it fits comfortably against the wall.

The rug is centrally placed under the desk, oriented parallel to the room's dimensions. This placement enhances the aesthetic appeal and provides a cohesive look to the workspace. The rug's dimensions are 2.0 meters by 1.5 meters by 0.01 meters, fitting comfortably under the desk area without overlapping other objects.

A conflict was identified with the desk lamp initially intended to be placed on the desk. The width of the desk was too small to accommodate the lamp without causing spatial issues. To resolve this, the desk lamp was removed, as it was deemed less critical compared to the other elements specified by the user, ensuring the room remains functional and aesthetically pleasing.

Figure 10: Chain-of-Thought for user preference analysis, functional zone partition, object recommendation and part of object placement.

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For rug_1
- parent object: swivel_chair_1
  - calculation_steps:
    1. reason: Calculate size constraint for 'under' relation
       - calculation:
         - rug_1 size: 2.0x1.5x0.01
         - Cluster size (under): 0.0 (non-directional)
       - conclusion: No directional constraint applied
    2. reason: Calculate possible positions based on 'middle of the room' constraint
       - calculation:
         - x_min = x_max = 2.5
         - y_min = y_max = 2.5
         - z_min = z_max = 0.005
       - conclusion: Possible position: (2.5, 2.5, 2.5, 2.5, 0.005, 0.005)
    3. reason: Adjust for 'under desk_1' constraint
       - calculation:
         - x_min = max(2.5, 2.0426 - 1.8/2 - 2.0/2) = 1.0
         - y_min = max(2.5, 4.55 - 0.9/2 - 1.5/2) = 3.25
       - conclusion: Final position: x: 1.0, y: 3.25, z: 0.005
    4. reason: Collision check with swivel_chair_1
       - calculation:
         - Overlap detection: 1.0 ≤ 1.9482 ≤ 4.0 → No collision
       - conclusion: No collision detected
    5. reason: Final position calculation
       - calculation:
         - Selected position within overlap: x=1.8337, y=3.9774, z=0.005
       - conclusion: Final position: x: 1.8337, y: 3.9774, z: 0.005

For filing_cabinet_1
- calculation_steps:
  1. reason: Calculate rotation difference with other objects
     - calculation:
       - No other objects in proximity
     - conclusion: No rotation difference applicable
  2. reason: Calculate size constraint for 'west_wall' relation
     - calculation:
       - filing_cabinet_1 size: 0.8x0.6x1.8
       - Cluster size (west_wall): 0.0 (non-directional)
     - conclusion: No directional constraint applied
  3. reason: Calculate possible positions based on 'west_wall' constraint
     - calculation:
       - filing_cabinet_1 size: length=0.8, width=0.6, height=1.8
       - Room size: 5.0x5.0x3.0
       - x_min = 0 + 0.6/2 = 0.3
       - x_max = 0 + 0.6/2 = 0.3
       - y_min = 2.5 - 5.0/2 + 0.8/2 = 0.4
       - y_max = 2.5 + 5.0/2 - 0.8/2 = 4.6
       - z_min = z_max = 1.8/2 = 0.9
     - conclusion: Possible position: (0.3, 0.3, 0.4, 4.6, 0.9, 0.9)
  4. reason: Adjust boundaries for valid placement
     - calculation:
       - Adjusted cluster constraint: x(0.3-0.3), y(0.4-4.6)
       - Final coordinates: x=0.3, y=1.5713, z=0.9
     - conclusion: Final position: x: 0.3, y: 1.5713, z: 0.9
  5. reason: Collision check with other objects
     - calculation:
       - No other objects in proximity
     - conclusion: No collision detected
  6. reason: Final position calculation
     - calculation:
       - Selected position within overlap: x=0.3, y=1.5713, z=0.9
     - conclusion: Final position: x: 0.3, y: 1.5713, z: 0.9
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Figure 11: Part of Chain-of-Thought for determining the specific positions of objects in object placement.

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